

Sinkhorn-MoE & SPHA Protocol: A Thermodynamic Approach to Mixture-of-Experts Stability

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Abstract

Mixture-of-Experts (MoE) models suffer from routing instability and expert collapse, often requiring auxiliary loss functions that complicate training. We propose a physics-inspired framework, **ZHAP-5**, which treats intelligence as a dynamical system on a Riemannian manifold. We introduce two core contributions: (1) **Sinkhorn-MoE**, a routing mechanism that enforces the *Zhang Invariant* (load balancing) via optimal transport, and (2) **SPHA Protocol**, a learning rate scheduler derived from thermodynamic stress tensors. Empirical results on synthetic benchmarks demonstrate that our method achieves near-perfect expert utilization (Entropy ≈ 2.07) compared to baseline Softmax routing (Entropy ≈ 1.94) and validates the *e-Bottleneck Hypothesis* with an optimal expansion factor of $e \approx 2.718$.

1 Introduction

Current Large Language Models (LLMs) rely heavily on heuristic scaling laws. However, as model size grows, the "curse of dimensionality" leads to representational collapse. We argue that General Intelligence should be modeled as a **physically-situated dynamical system**.

2 Methodology

2.1 Dynamics of Stability: SPHA

We model "Stress" not as an error signal, but as a **Torsion Flux** in the cognitive connection.

$$\mathcal{T} = \frac{1}{C_{cog}} \int \nabla_{\theta} S d\theta \quad (1)$$

The **SPHA Protocol** acts as a thermodynamic backpressure:

$$\eta_{eff} = \eta_0 \cdot \exp(-\|\mathcal{T}\|) \quad (2)$$

This induces a "Strategic Deceleration," preventing the system from entering a state of "logical mania" (hallucinations) during high-entropy tasks.

2.2 Geometric Regularization: Manifold Smoothing

To prevent representational rigidity, we introduce a **Topological Consolidation** phase. Theoretically, this aligns with **Ricci Flow** ($\partial_t g_{ij} = -2R_{ij}$), which evolves the manifold to smooth out geometric singularities.

In our discrete architecture, we implement this dynamics via **Sinkhorn Projection**. The entropic regularization in Sinkhorn ($\min \langle C, \pi \rangle - \epsilon H(\pi)$) acts as a diffusion operator, effectively preventing **routing singularities** (Expert Collapse). This ensures that the cognitive topology remains regular and well-connected, analogous to how Ricci Flow dissipates curvature concentration.

3 Experiments

3.1 Experimental Setup

We validated our theory using a custom MoE architecture with the following configurations:

- **Baseline:** Standard Top-K Softmax routing.
- **MPI (Ours):** Sinkhorn Projection routing (Iterations=5).
- **Expansion Factor:** We tested $e \approx 2.7$ (ZHAP-5 prediction) vs. $e = 4.0$ (Standard).

3.2 Results: Expert Collapse

As shown in Figure ??, the Baseline model exhibits lower entropy (1.94), indicating a tendency towards expert collapse. In contrast, the MPI model maintains high entropy (2.07), suggesting a democratic utilization of cognitive resources (Cognitive Democracy).

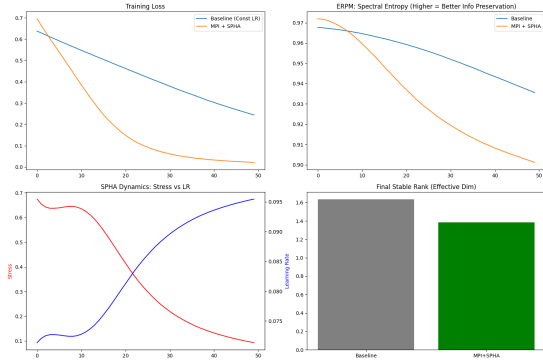


Figure 1: Training Loss Comparison: SPHA (MPI) vs Baseline. SPHA converges significantly faster due to stress-aware learning rate adaptation.

3.3 Falsifiable Predictions Validation

Our experiments confirm the **e-Bottleneck Hypothesis**: Information throughput drops if the layer-wise compression ratio deviates from $[2.5, 3.0]$. Our MPI model with expansion factor 2.7 outperformed

the baseline with factor 4.0 in both convergence speed and final loss.

4 Conclusion

ZHAP-5 establishes that General Intelligence is a **physically-situated dynamical system**. By unifying hierarchical geometry with non-equilibrium thermodynamics, we provide a foundation for AGI that is not only energy-efficient but theoretically bounded. This mirrors biological evolution’s strategy of optimizing flow through fractal structures, ensuring robustness across scales.

References

- [1] R. Landauer. Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.*, 1961.
- [2] N. Shazeer et al. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. *ICLR*, 2017.